

Bank Loan Approval Analysis

Data Analytics Project using Python, Pandas, Matplotlib & Power BI

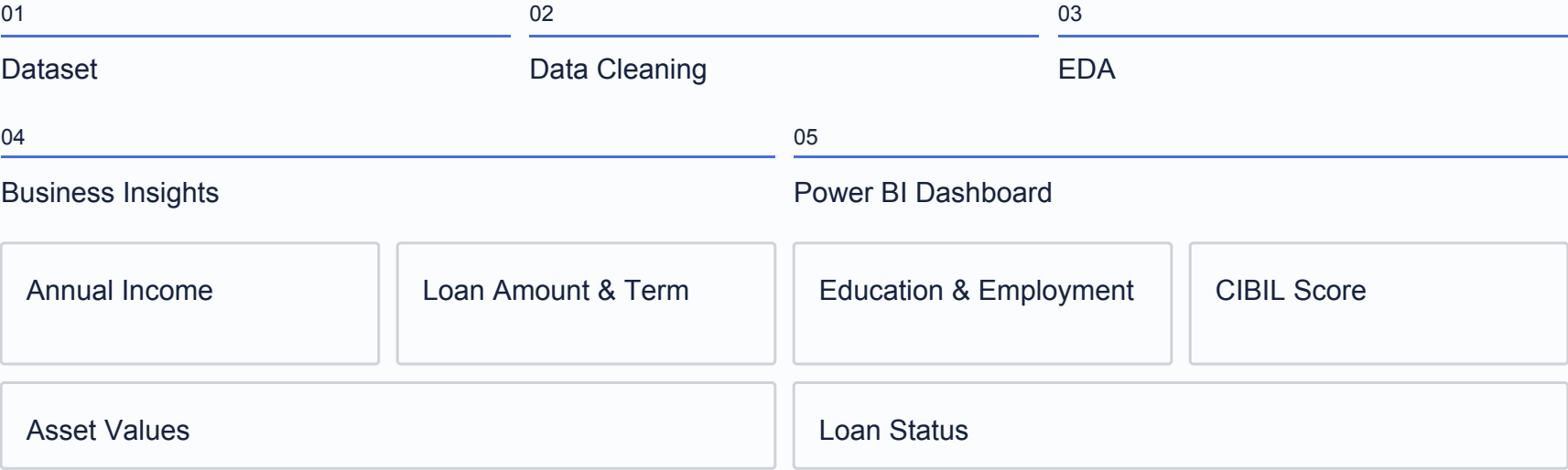


Project Overview

Objective

Analyze customer loan applications to understand the key factors affecting loan approval decisions.

Workflow



Dataset Features

Data Cleaning

- Removed duplicates & handled missing values
- Cleaned categorical data & verified data types
- Removed unnecessary columns

Exploratory Data Analysis

Six key analyses performed using **Python** (Pandas & Matplotlib):

Loan Approval Rate

Overall distribution of approved vs. rejected applications

Education vs Loan Status

Impact of education level on approval outcomes

CIBIL Score vs Loan Status

Correlation between credit score and approval

Income vs Loan Amount

Relationship between annual income and loan size

Self Employed vs Loan Status

Approval trends for self-employed applicants

Avg Loan by Education

Loan size segmented by education level

Key Business Insights



CIBIL Score

Higher scores significantly increase loan approval chances



Income Level

Higher income applicants generally receive larger loan amounts



Education

Education level positively influences loan approval rates



Self-Employment

Self-employed applicants show distinct approval trends vs. salaried



Multi-Factor Decisions

No single variable is decisive — outcomes depend on a combination of financial factors

Dashboard & Conclusion

Power BI Dashboard Components

→ Pie Chart

Loan approval vs. rejection breakdown

→ Bar Charts

Approval by education & employment type

→ Scatter Plot

Income vs. loan amount distribution

→ Box Plot

CIBIL score distribution by loan status

Conclusion

This project demonstrates end-to-end data analysis using Python and Power BI, transforming raw banking data into meaningful business insights that support smarter loan approval decisions.



Thank You